

Sleep Project Final Report

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1. Introduction

(paste intro here)

2. Methods

(paste methods here)

Step by Step Explanation (add if wanted):

- **Set up** *(add if wanted)*

3. Results

- **Descriptive Results**
 - Through our upper quartile study on sleep quality we discovered that the top-quartile sleepers had a much longer sleep duration as well as lower caffeine use and less device time compared to sleepers who were not in the top-quartile. Unexpectedly, top sleepers also had higher anxiety ratings, more social media use, and less physical activity.

- **Regression Results**

- The regression study was based on the independent variables: onset time, offset time, onset variability, offset variability, sleep duration and the dependent variables: alertness, anxiety, wake difficulty, and sleep quality. Due to the recurring problem of heteroscedasticity violating the assumption of homoscedasticity in the linear regression analyses conducted for each dependent variable we used an HC3 heteroskedasticity-robust standard errors test to interpret coefficients more accurately. The results showed that sleep duration is a significant predictor for alertness ($p = 0.0096$, $\beta = 0.0023$), indicating that longer sleep duration is associated with higher alertness.

Warning: package 'readxl' was built under R version 4.4.1

Warning: package 'dplyr' was built under R version 4.4.3

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Warning: There was 1 warning in `summarise()`.

i In argument: `across(where(is.numeric), mean, na.rm = TRUE)`.

i In group 1: `` step1\$`Global Participant ID` = "PUSH\_102" ``.

Caused by warning:

! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.

Supply arguments directly to `fns` through an anonymous function instead.

Previously

across(a:b, mean, na.rm = TRUE)

Now

across(a:b, \(\x\) mean(x, na.rm = TRUE))

Warning: package 'openxlsx' was built under R version 4.4.1

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'car' was built under R version 4.4.3

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

Warning: package 'corrplot' was built under R version 4.4.1

corrplot 0.95 loaded

Warning: package 'lmtest' was built under R version 4.4.1

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.5158	-1.3879	-0.0531	1.8536	4.8284

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

(Intercept)	3.74865	1.15923	3.234	0.00152 **
`Sleep Offset Time_Decimal Hour`	0.14614	0.12888	1.134	0.25869
`Sleep Onset Decimal Hour`	0.06177	0.02190	2.821	0.00547 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.549 on 144 degrees of freedom

(25 observations deleted due to missingness)

Multiple R-squared: 0.05283, Adjusted R-squared: 0.03968

F-statistic: 4.016 on 2 and 144 DF, p-value: 0.02008

studentized Breusch-Pagan test

data: alertness1

BP = 7.1558, df = 2, p-value = 0.02793

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.9742	-1.8143	-0.1123	1.7018	4.7731

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.6432271	1.1377990	3.202	0.00168 **
`Sleep Duration`	0.0021297	0.0008247	2.582	0.01082 *
`Sleep Offset Time_Decimal Hour`	0.0542421	0.1313257	0.413	0.68020
`Sleep Onset Decimal Hour`	0.0417437	0.0228385	1.828	0.06967 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.501 on 143 degrees of freedom

(25 observations deleted due to missingness)

Multiple R-squared: 0.09503, Adjusted R-squared: 0.07605

F-statistic: 5.006 on 3 and 143 DF, p-value: 0.00249

studentized Breusch-Pagan test

```
data: alertness2
BP = 11.42, df = 3, p-value = 0.009658
```

```
Call:
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

```
Residuals:
      Min       1Q   Median       3Q      Max
-4.6906 -1.5393  0.0393  1.6834  5.1780
```

```
Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                   3.7175013   1.2809727   2.902  0.00431 **
`Sleep Duration`               0.0022772   0.0008369   2.721  0.00734 **
`Sleep Offset Time_Decimal Hour` 0.0353955   0.1320510   0.268  0.78906
`Sleep Onset Decimal Hour`      0.0382507   0.0239230   1.599  0.11211
onset_var                     0.0585856   0.0449249   1.304  0.19436
offset_var                    -0.2821057   0.2730019  -1.033  0.30324
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.486 on 139 degrees of freedom
```

```
(27 observations deleted due to missingness)
```

```
Multiple R-squared:  0.1135,    Adjusted R-squared:  0.08165
```

```
F-statistic:  3.56 on 5 and 139 DF,  p-value: 0.004636
```

```
studentized Breusch-Pagan test
```

```
data: alertness3
BP = 12.35, df = 5, p-value = 0.0303
```

```
Call:
lm(formula = `Wake Difficulty Rating` ~ `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour`, data = step1)
```

```
Residuals:
      Min       1Q   Median       3Q      Max
```

-5.6511 -2.1743 0.2513 2.4092 4.8218

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.14781	1.25224	2.514	0.0131 *
`Sleep Offset Time_Decimal Hour`	0.29097	0.13897	2.094	0.0381 *
`Sleep Onset Decimal Hour`	0.05974	0.02384	2.506	0.0133 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.747 on 141 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.05501, Adjusted R-squared: 0.0416

F-statistic: 4.104 on 2 and 141 DF, p-value: 0.01852

studentized Breusch-Pagan test

data: wake_diff1

BP = 6.8965, df = 2, p-value = 0.0318

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.414	-2.117	0.225	2.260	4.850

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1181202	1.2547764	2.485	0.0141 *
`Sleep Duration`	0.0006999	0.0009274	0.755	0.4517
`Sleep Offset Time_Decimal Hour`	0.2601502	0.1450513	1.794	0.0751 .
`Sleep Onset Decimal Hour`	0.0530960	0.0254440	2.087	0.0387 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.751 on 140 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.05884, Adjusted R-squared: 0.03867

F-statistic: 2.917 on 3 and 140 DF, p-value: 0.03641

studentized Breusch-Pagan test

data: wake_diff2

BP = 12.6, df = 3, p-value = 0.005587

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.3187	-2.1928	0.3314	2.0396	4.9466

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.992346	1.400665	3.564	0.000504 ***
`Sleep Duration`	0.001123	0.000929	1.209	0.228878
`Sleep Offset Time_Decimal Hour`	0.217096	0.144162	1.506	0.134408
`Sleep Onset Decimal Hour`	0.031007	0.026295	1.179	0.240379
onset_var	-0.064145	0.049466	-1.297	0.196913
offset_var	-0.719649	0.300054	-2.398	0.017824 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.703 on 136 degrees of freedom

(30 observations deleted due to missingness)

Multiple R-squared: 0.1076, Adjusted R-squared: 0.07479

F-statistic: 3.28 on 5 and 136 DF, p-value: 0.007913

studentized Breusch-Pagan test

data: wake_diff3

BP = 17.556, df = 5, p-value = 0.003558

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.4714	-0.4869	1.1238	1.4998	1.6805

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.15195	1.04755	7.782	1.3e-12 ***
`Sleep Offset Time_Decimal Hour`	0.07706	0.11646	0.662	0.509
`Sleep Onset Decimal Hour`	-0.00936	0.01979	-0.473	0.637

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.302 on 143 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.006549, Adjusted R-squared: -0.007346

F-statistic: 0.4713 on 2 and 143 DF, p-value: 0.6251

studentized Breusch-Pagan test

data: anxiety1

BP = 1.1364, df = 2, p-value = 0.5666

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.1469	-0.7049	1.0838	1.3818	1.9541

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.0925675	1.0450930	7.743	1.66e-12 ***
`Sleep Duration`	0.0010575	0.0007641	1.384	0.169
`Sleep Offset Time_Decimal Hour`	0.0321888	0.1205281	0.267	0.790
`Sleep Onset Decimal Hour`	-0.0194257	0.0210236	-0.924	0.357

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.295 on 142 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.01977, Adjusted R-squared: -0.0009389

F-statistic: 0.9547 on 3 and 142 DF, p-value: 0.4161

studentized Breusch-Pagan test

data: anxiety2

BP = 9.4659, df = 3, p-value = 0.0237

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.9946	-0.6792	1.0401	1.4167	2.1795

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.3598911	1.1990470	6.972	1.18e-10 ***
`Sleep Duration`	0.0011577	0.0007859	1.473	0.143
`Sleep Offset Time_Decimal Hour`	0.0354732	0.1231159	0.288	0.774
`Sleep Onset Decimal Hour`	-0.0248727	0.0223228	-1.114	0.267
onset_var	-0.0309295	0.0420798	-0.735	0.464
offset_var	-0.0512598	0.2551679	-0.201	0.841

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.318 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.02573, Adjusted R-squared: -0.009565

F-statistic: 0.729 on 5 and 138 DF, p-value: 0.6028

studentized Breusch-Pagan test

data: anxiety3

BP = 10.383, df = 5, p-value = 0.06509

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.4794	-1.0730	0.3397	1.4783	3.0204

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.78792	0.92339	6.268	4.06e-09 ***
`Sleep Offset Time_Decimal Hour`	0.14209	0.10286	1.381	0.1693
`Sleep Onset Decimal Hour`	0.03639	0.01751	2.078	0.0395 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.03 on 143 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.03344, Adjusted R-squared: 0.01992

F-statistic: 2.473 on 2 and 143 DF, p-value: 0.0879

studentized Breusch-Pagan test

data: sleep_quality1

BP = 1.2834, df = 2, p-value = 0.5264

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.4114	-1.0942	0.2993	1.4779	3.0184

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.7779106	0.9269421	6.233	4.91e-09 ***

`Sleep Duration`	0.0002020	0.0006718	0.301	0.7641
`Sleep Offset Time_Decimal Hour`	0.1333790	0.1071760	1.244	0.2154
`Sleep Onset Decimal Hour`	0.0344898	0.0186702	1.847	0.0668 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.037 on 142 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.03405, Adjusted R-squared: 0.01364

F-statistic: 1.669 on 3 and 142 DF, p-value: 0.1765

studentized Breusch-Pagan test

data: sleep_quality2

BP = 4.0417, df = 3, p-value = 0.257

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.4510	-1.0031	0.3062	1.4986	3.0138

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.8921655	1.0638612	5.538	1.49e-07 ***
`Sleep Duration`	0.0002554	0.0006951	0.367	0.714
`Sleep Offset Time_Decimal Hour`	0.1343439	0.1099508	1.222	0.224
`Sleep Onset Decimal Hour`	0.0320627	0.0199638	1.606	0.111
onset_var	-0.0102492	0.0376939	-0.272	0.786
offset_var	-0.0367834	0.2268967	-0.162	0.871

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.064 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.03469, Adjusted R-squared: -0.0002878

F-statistic: 0.9918 on 5 and 138 DF, p-value: 0.4252

studentized Breusch-Pagan test

data: sleep_quality3

BP = 13.291, df = 5, p-value = 0.0208

`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.204114	1.208512
`Sleep Onset Decimal Hour`	onset_var
1.376575	1.053563
offset_var	
1.078269	

Warning: package 'Hmisc' was built under R version 4.4.3

Attaching package: 'Hmisc'

The following objects are masked from 'package:dplyr':

src, summarize

The following objects are masked from 'package:base':

format.pval, units

Registered S3 method overwritten by 'GGally':

method from
+.gg ggplot2

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	1.00000000	0.42018413	0.03308203
Wake Difficulty Rating	0.42018413	1.00000000	0.03607769
Anxiety Rating	0.03308203	0.03607769	1.00000000
Sleep Quality	0.30883513	0.15515198	0.29848043

	Sleep Quality
Alertness Rating	0.3088351
Wake Difficulty Rating	0.1551520
Anxiety Rating	0.2984804
Sleep Quality	1.0000000

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	1.00	0.42	0.03
Wake Difficulty Rating	0.42	1.00	0.04
Anxiety Rating	0.03	0.04	1.00
Sleep Quality	0.31	0.16	0.30

	Sleep Quality
Alertness Rating	0.31
Wake Difficulty Rating	0.16
Anxiety Rating	0.30
Sleep Quality	1.00

n

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	149	146	148
Wake Difficulty Rating	146	146	145
Anxiety Rating	148	145	148
Sleep Quality	148	145	147

	Sleep Quality
Alertness Rating	148
Wake Difficulty Rating	145
Anxiety Rating	147
Sleep Quality	148

P

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	0.0000	0.6898	
Wake Difficulty Rating	0.0000	0.6666	
Anxiety Rating	0.6898	0.6666	
Sleep Quality	0.0001	0.0624	0.0002

	Sleep Quality
Alertness Rating	0.0001
Wake Difficulty Rating	0.0624
Anxiety Rating	0.0002
Sleep Quality	

Warning: package 'sandwich' was built under R version 4.4.1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.71750126	1.20380995	3.0881	0.002433 **

`Sleep Duration`	0.00227718	0.00087143	2.6131	0.009959	**
`Sleep Offset Time_Decimal Hour`	0.03539554	0.13325336	0.2656	0.790921	
`Sleep Onset Decimal Hour`	0.03825066	0.02602504	1.4698	0.143886	
onset_var	0.05858558	0.05074411	1.1545	0.250264	
offset_var	-0.28210566	0.22575268	-1.2496	0.213537	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.35989111	1.67780188	4.9826	1.847e-06	***
`Sleep Duration`	0.00115773	0.00086787	1.3340	0.1844	
`Sleep Offset Time_Decimal Hour`	0.03547316	0.17240382	0.2058	0.8373	
`Sleep Onset Decimal Hour`	-0.02487272	0.02642234	-0.9414	0.3482	
onset_var	-0.03092954	0.04037443	-0.7661	0.4449	
offset_var	-0.05125984	0.32376833	-0.1583	0.8744	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.99234644	1.32378744	3.7713	0.0002414	***
`Sleep Duration`	0.00112287	0.00087052	1.2899	0.1992770	
`Sleep Offset Time_Decimal Hour`	0.21709630	0.14366658	1.5111	0.1330800	
`Sleep Onset Decimal Hour`	0.03100672	0.02764274	1.1217	0.2639687	
onset_var	-0.06414519	0.05279502	-1.2150	0.2264768	
offset_var	-0.71964942	0.25775204	-2.7920	0.0059928	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.89216552	1.11874964	5.2667	5.206e-07	***
`Sleep Duration`	0.00025542	0.00070248	0.3636	0.7167	
`Sleep Offset Time_Decimal Hour`	0.13434388	0.11047080	1.2161	0.2260	
`Sleep Onset Decimal Hour`	0.03206272	0.02098630	1.5278	0.1289	

```
onset_var          -0.01024923  0.04173218 -0.2456    0.8064
offset_var         -0.03678338  0.18998515 -0.1936    0.8468
---
```

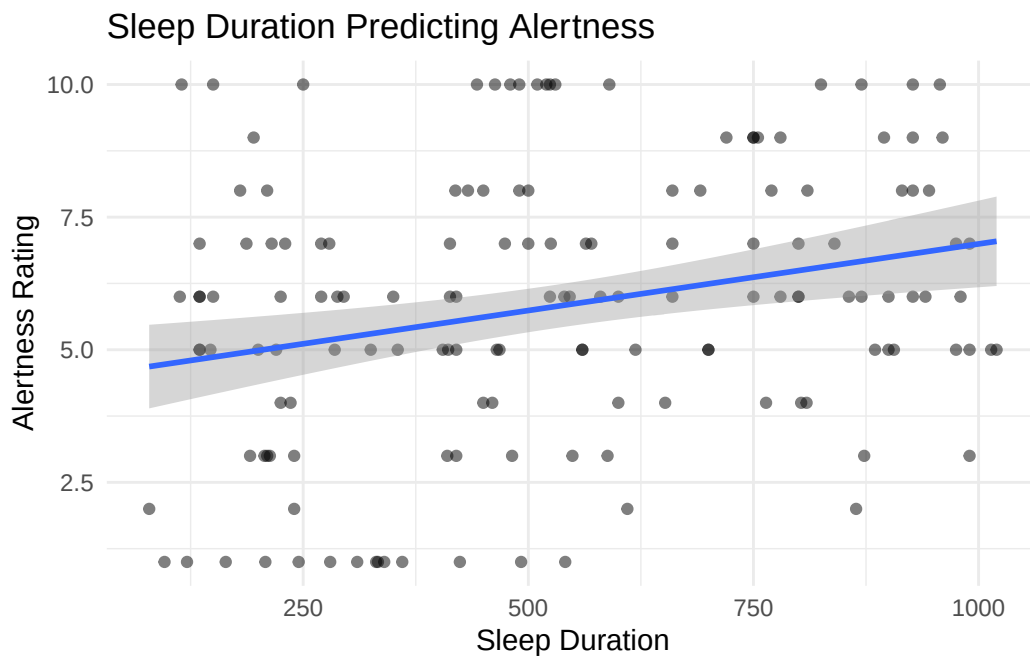
```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Warning: package 'broom' was built under R version 4.4.3
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 24 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 24 rows containing missing values or values outside the scale range
(`geom_point()`).
```



- Another notable result is offset variability as a significant predictor for wake difficu.

```
::: {.cell}
```

```
::: {.cell-output .cell-output-stdout}
```

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.5158	-1.3879	-0.0531	1.8536	4.8284

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.74865	1.15923	3.234	0.00152 **
`Sleep Offset Time_Decimal Hour`	0.14614	0.12888	1.134	0.25869
`Sleep Onset Decimal Hour`	0.06177	0.02190	2.821	0.00547 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.549 on 144 degrees of freedom

(25 observations deleted due to missingness)

Multiple R-squared: 0.05283, Adjusted R-squared: 0.03968

F-statistic: 4.016 on 2 and 144 DF, p-value: 0.02008

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: alertness1

BP = 7.1558, df = 2, p-value = 0.02793

:::

::: {.cell-output .cell-output-stdout}

```
Call:
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.9742	-1.8143	-0.1123	1.7018	4.7731

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.6432271	1.1377990	3.202	0.00168 **
`Sleep Duration`	0.0021297	0.0008247	2.582	0.01082 *
`Sleep Offset Time_Decimal Hour`	0.0542421	0.1313257	0.413	0.68020
`Sleep Onset Decimal Hour`	0.0417437	0.0228385	1.828	0.06967 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.501 on 143 degrees of freedom

(25 observations deleted due to missingness)

Multiple R-squared: 0.09503, Adjusted R-squared: 0.07605

F-statistic: 5.006 on 3 and 143 DF, p-value: 0.00249

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: alertness2

BP = 11.42, df = 3, p-value = 0.009658

:::

::: {.cell-output .cell-output-stdout}

```
Call:
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.6906	-1.5393	0.0393	1.6834	5.1780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.7175013	1.2809727	2.902	0.00431	**
`Sleep Duration`	0.0022772	0.0008369	2.721	0.00734	**
`Sleep Offset Time_Decimal Hour`	0.0353955	0.1320510	0.268	0.78906	
`Sleep Onset Decimal Hour`	0.0382507	0.0239230	1.599	0.11211	
onset_var	0.0585856	0.0449249	1.304	0.19436	
offset_var	-0.2821057	0.2730019	-1.033	0.30324	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.486 on 139 degrees of freedom

(27 observations deleted due to missingness)

Multiple R-squared: 0.1135, Adjusted R-squared: 0.08165

F-statistic: 3.56 on 5 and 139 DF, p-value: 0.004636

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: alertness3

BP = 12.35, df = 5, p-value = 0.0303

:::

::: {.cell-output .cell-output-stdout}

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.6511	-2.1743	0.2513	2.4092	4.8218

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.14781	1.25224	2.514	0.0131 *
`Sleep Offset Time_Decimal Hour`	0.29097	0.13897	2.094	0.0381 *
`Sleep Onset Decimal Hour`	0.05974	0.02384	2.506	0.0133 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.747 on 141 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.05501, Adjusted R-squared: 0.0416

F-statistic: 4.104 on 2 and 141 DF, p-value: 0.01852

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: wake_diff1

BP = 6.8965, df = 2, p-value = 0.0318

:::

::: {.cell-output .cell-output-stdout}

...

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.414	-2.117	0.225	2.260	4.850

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1181202	1.2547764	2.485	0.0141 *
`Sleep Duration`	0.0006999	0.0009274	0.755	0.4517
`Sleep Offset Time_Decimal Hour`	0.2601502	0.1450513	1.794	0.0751 .
`Sleep Onset Decimal Hour`	0.0530960	0.0254440	2.087	0.0387 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.751 on 140 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.05884, Adjusted R-squared: 0.03867

F-statistic: 2.917 on 3 and 140 DF, p-value: 0.03641

...

:::

::: {.cell-output .cell-output-stdout}

...

studentized Breusch-Pagan test

data: wake_diff2

BP = 12.6, df = 3, p-value = 0.005587

...

:::

::: {.cell-output .cell-output-stdout}

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.3187	-2.1928	0.3314	2.0396	4.9466

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.992346	1.400665	3.564	0.000504 ***
`Sleep Duration`	0.001123	0.000929	1.209	0.228878
`Sleep Offset Time_Decimal Hour`	0.217096	0.144162	1.506	0.134408
`Sleep Onset Decimal Hour`	0.031007	0.026295	1.179	0.240379
onset_var	-0.064145	0.049466	-1.297	0.196913
offset_var	-0.719649	0.300054	-2.398	0.017824 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.703 on 136 degrees of freedom

(30 observations deleted due to missingness)

Multiple R-squared: 0.1076, Adjusted R-squared: 0.07479

F-statistic: 3.28 on 5 and 136 DF, p-value: 0.007913

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: wake_diff3

BP = 17.556, df = 5, p-value = 0.003558

:::

```

::: {.cell-output .cell-output-stdout}
...

Call:
lm(formula = `Anxiety Rating` ~ `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`, data = step1)

Residuals:
    Min       1Q   Median       3Q      Max
-7.4714 -0.4869  1.1238  1.4998  1.6805

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)         8.15195     1.04755   7.782 1.3e-12 ***
`Sleep Offset Time_Decimal Hour`  0.07706     0.11646   0.662  0.509
`Sleep Onset Decimal Hour`      -0.00936     0.01979  -0.473  0.637
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.302 on 143 degrees of freedom
(26 observations deleted due to missingness)
Multiple R-squared:  0.006549, Adjusted R-squared:  -0.007346
F-statistic: 0.4713 on 2 and 143 DF,  p-value: 0.6251
...

:::

::: {.cell-output .cell-output-stdout}
...

    studentized Breusch-Pagan test

data:  anxiety1
BP = 1.1364, df = 2, p-value = 0.5666
...

:::

::: {.cell-output .cell-output-stdout}

```

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.1469	-0.7049	1.0838	1.3818	1.9541

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.0925675	1.0450930	7.743	1.66e-12 ***
`Sleep Duration`	0.0010575	0.0007641	1.384	0.169
`Sleep Offset Time_Decimal Hour`	0.0321888	0.1205281	0.267	0.790
`Sleep Onset Decimal Hour`	-0.0194257	0.0210236	-0.924	0.357

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.295 on 142 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.01977, Adjusted R-squared: -0.0009389

F-statistic: 0.9547 on 3 and 142 DF, p-value: 0.4161

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: anxiety2

BP = 9.4659, df = 3, p-value = 0.0237

:::

::: {.cell-output .cell-output-stdout}

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.9946	-0.6792	1.0401	1.4167	2.1795

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.3598911	1.1990470	6.972	1.18e-10 ***
`Sleep Duration`	0.0011577	0.0007859	1.473	0.143
`Sleep Offset Time_Decimal Hour`	0.0354732	0.1231159	0.288	0.774
`Sleep Onset Decimal Hour`	-0.0248727	0.0223228	-1.114	0.267
onset_var	-0.0309295	0.0420798	-0.735	0.464
offset_var	-0.0512598	0.2551679	-0.201	0.841

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.318 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.02573, Adjusted R-squared: -0.009565

F-statistic: 0.729 on 5 and 138 DF, p-value: 0.6028

:::

::: {.cell-output .cell-output-stdout}

studentized Breusch-Pagan test

data: anxiety3

BP = 10.383, df = 5, p-value = 0.06509

:::


```

::: {.cell-output .cell-output-stdout}
...

Call:
lm(formula = `Sleep Quality` ~ `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`, data = step1)

Residuals:
    Min       1Q   Median       3Q      Max
-6.4794 -1.0730  0.3397  1.4783  3.0204

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                   5.78792     0.92339   6.268 4.06e-09 ***
`Sleep Offset Time_Decimal Hour` 0.14209     0.10286   1.381  0.1693
`Sleep Onset Decimal Hour`       0.03639     0.01751   2.078  0.0395 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.03 on 143 degrees of freedom
(26 observations deleted due to missingness)
Multiple R-squared:  0.03344,    Adjusted R-squared:  0.01992
F-statistic: 2.473 on 2 and 143 DF,  p-value: 0.0879
...

:::

::: {.cell-output .cell-output-stdout}
...

      studentized Breusch-Pagan test

data:  sleep_quality1
BP = 1.2834, df = 2, p-value = 0.5264
...

:::

```

```

::: {.cell-output .cell-output-stdout}
...

Call:
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`, data = step1)

Residuals:
    Min       1Q   Median       3Q      Max
-6.4114 -1.0942  0.2993  1.4779  3.0184

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      5.7779106   0.9269421   6.233 4.91e-09 ***
`Sleep Duration`  0.0002020   0.0006718   0.301  0.7641
`Sleep Offset Time_Decimal Hour` 0.1333790   0.1071760   1.244  0.2154
`Sleep Onset Decimal Hour`      0.0344898   0.0186702   1.847  0.0668 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.037 on 142 degrees of freedom
(26 observations deleted due to missingness)
Multiple R-squared:  0.03405,    Adjusted R-squared:  0.01364
F-statistic: 1.669 on 3 and 142 DF,  p-value: 0.1765
...

:::

::: {.cell-output .cell-output-stdout}
...

    studentized Breusch-Pagan test

data:  sleep_quality2
BP = 4.0417, df = 3, p-value = 0.257
...

:::

```

```

::: {.cell-output .cell-output-stdout}
...

Call:
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)

Residuals:
    Min       1Q   Median       3Q      Max
-6.4510 -1.0031  0.3062  1.4986  3.0138

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      5.8921655   1.0638612    5.538 1.49e-07 ***
`Sleep Duration`  0.0002554   0.0006951    0.367   0.714
`Sleep Offset Time_Decimal Hour` 0.1343439   0.1099508    1.222   0.224
`Sleep Onset Decimal Hour`      0.0320627   0.0199638    1.606   0.111
onset_var        -0.0102492   0.0376939   -0.272   0.786
offset_var       -0.0367834   0.2268967   -0.162   0.871
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.064 on 138 degrees of freedom
(28 observations deleted due to missingness)
Multiple R-squared:  0.03469,    Adjusted R-squared:  -0.0002878
F-statistic: 0.9918 on 5 and 138 DF,  p-value: 0.4252
...

:::

::: {.cell-output .cell-output-stdout}
...

      studentized Breusch-Pagan test

data:  sleep_quality3
BP = 13.291, df = 5, p-value = 0.0208
...

```

:::

::: {.cell-output .cell-output-stdout}

...

`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.204114	1.208512
`Sleep Onset Decimal Hour`	onset_var
1.376575	1.053563
offset_var	
1.078269	

...

:::

::: {.cell-output .cell-output-stdout}

...

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	1.00000000	0.42018413	0.03308203
Wake Difficulty Rating	0.42018413	1.00000000	0.03607769
Anxiety Rating	0.03308203	0.03607769	1.00000000
Sleep Quality	0.30883513	0.15515198	0.29848043

	Sleep Quality
Alertness Rating	0.3088351
Wake Difficulty Rating	0.1551520
Anxiety Rating	0.2984804
Sleep Quality	1.0000000

...

:::

::: {.cell-output .cell-output-stdout}

...

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	1.00	0.42	0.03
Wake Difficulty Rating	0.42	1.00	0.04
Anxiety Rating	0.03	0.04	1.00
Sleep Quality	0.31	0.16	0.30

	Sleep Quality

Alertness Rating	0.31
Wake Difficulty Rating	0.16
Anxiety Rating	0.30
Sleep Quality	1.00

n

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	149	146	148
Wake Difficulty Rating	146	146	145
Anxiety Rating	148	145	148
Sleep Quality	148	145	147

	Sleep Quality
Alertness Rating	148
Wake Difficulty Rating	145
Anxiety Rating	147
Sleep Quality	148

P

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	0.0000	0.6898	
Wake Difficulty Rating	0.0000	0.6666	
Anxiety Rating	0.6898	0.6666	
Sleep Quality	0.0001	0.0624	0.0002

	Sleep Quality
Alertness Rating	0.0001
Wake Difficulty Rating	0.0624
Anxiety Rating	0.0002
Sleep Quality	

...

:::

::: {.cell-output .cell-output-stdout}

...

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.71750126	1.20380995	3.0881	0.002433 **
`Sleep Duration`	0.00227718	0.00087143	2.6131	0.009959 **
`Sleep Offset Time_Decimal Hour`	0.03539554	0.13325336	0.2656	0.790921

`Sleep Onset Decimal Hour`	0.03825066	0.02602504	1.4698	0.143886
onset_var	0.05858558	0.05074411	1.1545	0.250264
offset_var	-0.28210566	0.22575268	-1.2496	0.213537

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

:::

::: {.cell-output .cell-output-stdout}

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.35989111	1.67780188	4.9826	1.847e-06	***
`Sleep Duration`	0.00115773	0.00086787	1.3340	0.1844	
`Sleep Offset Time_Decimal Hour`	0.03547316	0.17240382	0.2058	0.8373	
`Sleep Onset Decimal Hour`	-0.02487272	0.02642234	-0.9414	0.3482	
onset_var	-0.03092954	0.04037443	-0.7661	0.4449	
offset_var	-0.05125984	0.32376833	-0.1583	0.8744	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

:::

::: {.cell-output .cell-output-stdout}

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.99234644	1.32378744	3.7713	0.0002414	***
`Sleep Duration`	0.00112287	0.00087052	1.2899	0.1992770	
`Sleep Offset Time_Decimal Hour`	0.21709630	0.14366658	1.5111	0.1330800	
`Sleep Onset Decimal Hour`	0.03100672	0.02764274	1.1217	0.2639687	
onset_var	-0.06414519	0.05279502	-1.2150	0.2264768	
offset_var	-0.71964942	0.25775204	-2.7920	0.0059928	**

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
---

:::

::: {.cell-output .cell-output-stdout}
---

t test of coefficients:

              Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.89216552  1.11874964  5.2667 5.206e-07 ***
`Sleep Duration` 0.00025542  0.00070248  0.3636  0.7167
`Sleep Offset Time_Decimal Hour` 0.13434388  0.11047080  1.2161  0.2260
`Sleep Onset Decimal Hour` 0.03206272  0.02098630  1.5278  0.1289
onset_var      -0.01024923  0.04173218 -0.2456  0.8064
offset_var     -0.03678338  0.18998515 -0.1936  0.8468
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
---

:::

::: {.cell-output .cell-output-stderr}
---

`geom_smooth()` using formula = 'y ~ x'
---

:::

::: {.cell-output .cell-output-stderr}
---

Warning: Removed 28 rows containing non-finite outside the scale range
(`stat_smooth()`).
---

```

```

:::

::: {.cell-output .cell-output-stderr}

```
Warning: Removed 28 rows containing missing values or values outside the scale range
(`geom_point()`).
```

:::

::: {.cell-output-display}
![] (Sleep-Project-Final-Report_files/figure-pdf/Offset Variability & Wake Difficulty-1.pdf)
:::
:::

```

- **Correlation Results**

- We also looked into correlation between the dependent variables and found significant correlations between alertness & wake difficulty ($r=0.42$), alertness & sleep quality ($r=0.31$), and anxiety & sleep quality ($r=0.30$).

- **Mixed Effects Results**

- The data included 147 nights from 27 participants, within the data we noticed some trends although not statistically significance still peaked our interest. The trends we found were the association of more sleep than usual to a better reported sleep quality as well as a later sleep onset associated with a worse reported sleep quality. Though these are interesting trends it's important to emphasize that no predictors reached statistical significance in this study. The model assumptions were mostly satisfied showing no major variance issues and although there were some slight left-skew residuals the overall model fit is considered acceptable.

- **Takeaways**

- Sleep duration is the strongest consistent predictor within our study. However, most other sleep variables showed limited predictive powers. Some relationships exist between key outcomes, but casual predictors are weak overall.

4. Conclusions/Future Work

(paste conclusion here)

5. Appendix

```
#ALL OF OUR DATA

#REGRESSION ANALYSIS
#Data Input ----
library(readxl)
source("./excel_date_time_utils.R")

step1 <- suppressWarnings(read_excel("./merged_diaries_actigraphy.xlsx", sheet = "best_clean"))

step1$`Sleep Onset Decimal Hour` <- as.numeric(step1$`Sleep Onset Decimal Hour`)
step1$`Sleep Offset Time_Decimal Hour` <- as.numeric(step1$`Sleep Offset Time_Decimal Hour`)

#making means for jules----
library(dplyr)

means <- step1 %>%
  group_by(step1$`Global Participant ID`) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))

#install.packages("openxlsx")
library(openxlsx)

write.xlsx(means, "mean_per_person.xlsx")

#All together vers----

#Building blocks----

#install.packages("dplyr")
#install.packages("ggplot2")
#install.packages("car")
#install.packages("corrplot")
#install.packages("lmtest")
library(dplyr)
library(ggplot2)
library(car)
```

```

library(corrplot)
library(dplyr)
library(lmtest)

#Variability Dataset

sleep_var <- step1 %>%
  group_by(`Global Participant ID`) %>%
  summarise(
    onset_var = sd(`Sleep Onset Decimal Hour`, na.rm = TRUE),
    offset_var = sd(`Sleep Offset Time_Decimal Hour`, na.rm = TRUE),
    mean_alertness = mean(`Alertness Rating`, na.rm = TRUE),
    mean_wake_diff = mean(`Wake Difficulty Rating`, na.rm = TRUE),
    mean_anxiety = mean(`Anxiety Rating`, na.rm = TRUE),
    mean_sleep_quality = mean(`Sleep Quality`, na.rm = TRUE)
  )

step1 <- merge(step1, sleep_var,
               by = "Global Participant ID")

#Alertness Analysis----

#onset & offset
alertness1 <- lm(
  `Alertness Rating` ~
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`,
  data = step1
)

summary(alertness1)

```

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.5158	-1.3879	-0.0531	1.8536	4.8284

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.74865	1.15923	3.234	0.00152 **
`Sleep Offset Time_Decimal Hour`	0.14614	0.12888	1.134	0.25869
`Sleep Onset Decimal Hour`	0.06177	0.02190	2.821	0.00547 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.549 on 144 degrees of freedom

(25 observations deleted due to missingness)

Multiple R-squared: 0.05283, Adjusted R-squared: 0.03968

F-statistic: 4.016 on 2 and 144 DF, p-value: 0.02008

```
bptest(alertness1)
```

studentized Breusch-Pagan test

data: alertness1

BP = 7.1558, df = 2, p-value = 0.02793

```
#onset, offset, duration
alertness2 <- lm(
  `Alertness Rating` ~
    `Sleep Duration` +
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`,
  data = step1
)

summary(alertness2)
```

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.9742	-1.8143	-0.1123	1.7018	4.7731

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.6432271	1.1377990	3.202	0.00168 **
`Sleep Duration`	0.0021297	0.0008247	2.582	0.01082 *
`Sleep Offset Time_Decimal Hour`	0.0542421	0.1313257	0.413	0.68020
`Sleep Onset Decimal Hour`	0.0417437	0.0228385	1.828	0.06967 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.501 on 143 degrees of freedom

(25 observations deleted due to missingness)

Multiple R-squared: 0.09503, Adjusted R-squared: 0.07605

F-statistic: 5.006 on 3 and 143 DF, p-value: 0.00249

```
bptest(alertness2)
```

studentized Breusch-Pagan test

data: alertness2

BP = 11.42, df = 3, p-value = 0.009658

```
#onset, offset, duration, variability
alertness3 <- lm(
  `Alertness Rating` ~
    `Sleep Duration` +
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour` +
    onset_var +
    offset_var,
  data = step1
)

summary(alertness3)
```

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-4.6906 -1.5393 0.0393 1.6834 5.1780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.7175013	1.2809727	2.902	0.00431	**
`Sleep Duration`	0.0022772	0.0008369	2.721	0.00734	**
`Sleep Offset Time_Decimal Hour`	0.0353955	0.1320510	0.268	0.78906	
`Sleep Onset Decimal Hour`	0.0382507	0.0239230	1.599	0.11211	
onset_var	0.0585856	0.0449249	1.304	0.19436	
offset_var	-0.2821057	0.2730019	-1.033	0.30324	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.486 on 139 degrees of freedom

(27 observations deleted due to missingness)

Multiple R-squared: 0.1135, Adjusted R-squared: 0.08165

F-statistic: 3.56 on 5 and 139 DF, p-value: 0.004636

```
bptest(alertness3)
```

studentized Breusch-Pagan test

data: alertness3

BP = 12.35, df = 5, p-value = 0.0303

```
#Wake Difficulty Analysis----

#onset & offset
wake_diff1 <- lm(
  `Wake Difficulty Rating` ~
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`,
  data = step1
)

summary(wake_diff1)
```

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Offset Time_Decimal Hour` +
```

```
`Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.6511	-2.1743	0.2513	2.4092	4.8218

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.14781	1.25224	2.514	0.0131 *
`Sleep Offset Time_Decimal Hour`	0.29097	0.13897	2.094	0.0381 *
`Sleep Onset Decimal Hour`	0.05974	0.02384	2.506	0.0133 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.747 on 141 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.05501, Adjusted R-squared: 0.0416

F-statistic: 4.104 on 2 and 141 DF, p-value: 0.01852

```
bptest(wake_diff1)
```

studentized Breusch-Pagan test

data: wake_diff1

BP = 6.8965, df = 2, p-value = 0.0318

```
#onset, offset, duration
wake_diff2 <- lm(
  `Wake Difficulty Rating` ~
    `Sleep Duration` +
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`,
  data = step1
)

summary(wake_diff2)
```

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour`
```

```

`Sleep Onset Decimal Hour`, data = step1)

Residuals:
    Min       1Q   Median       3Q      Max
-5.414 -2.117  0.225  2.260  4.850

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      3.1181202   1.2547764   2.485   0.0141 *
`Sleep Duration`  0.0006999   0.0009274   0.755   0.4517
`Sleep Offset Time_Decimal Hour` 0.2601502   0.1450513   1.794   0.0751 .
`Sleep Onset Decimal Hour`      0.0530960   0.0254440   2.087   0.0387 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.751 on 140 degrees of freedom
(28 observations deleted due to missingness)
Multiple R-squared:  0.05884,    Adjusted R-squared:  0.03867
F-statistic: 2.917 on 3 and 140 DF,  p-value: 0.03641

```

```
bptest(wake_diff2)
```

studentized Breusch-Pagan test

```

data:  wake_diff2
BP = 12.6, df = 3, p-value = 0.005587

```

```

#onset, offset, duration, variability
wake_diff3 <- lm(
  `Wake Difficulty Rating` ~
    `Sleep Duration` +
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour` +
    onset_var +
    offset_var,
  data = step1
)

summary(wake_diff3)

```

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.3187	-2.1928	0.3314	2.0396	4.9466

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.992346	1.400665	3.564	0.000504 ***
`Sleep Duration`	0.001123	0.000929	1.209	0.228878
`Sleep Offset Time_Decimal Hour`	0.217096	0.144162	1.506	0.134408
`Sleep Onset Decimal Hour`	0.031007	0.026295	1.179	0.240379
onset_var	-0.064145	0.049466	-1.297	0.196913
offset_var	-0.719649	0.300054	-2.398	0.017824 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.703 on 136 degrees of freedom

(30 observations deleted due to missingness)

Multiple R-squared: 0.1076, Adjusted R-squared: 0.07479

F-statistic: 3.28 on 5 and 136 DF, p-value: 0.007913

```
bptest(wake_diff3)
```

studentized Breusch-Pagan test

data: wake_diff3

BP = 17.556, df = 5, p-value = 0.003558

```
#Anxiety Analysis----
```

```
#onset & offset
```

```
anxiety1 <- lm(  
  `Anxiety Rating` ~  
    `Sleep Offset Time_Decimal Hour` +  
    `Sleep Onset Decimal Hour`,  
  data = step1  
)
```



```
summary(anxiety1)
```

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.4714	-0.4869	1.1238	1.4998	1.6805

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.15195	1.04755	7.782	1.3e-12 ***
`Sleep Offset Time_Decimal Hour`	0.07706	0.11646	0.662	0.509
`Sleep Onset Decimal Hour`	-0.00936	0.01979	-0.473	0.637

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.302 on 143 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.006549, Adjusted R-squared: -0.007346

F-statistic: 0.4713 on 2 and 143 DF, p-value: 0.6251

```
bptest(anxiety1)
```

studentized Breusch-Pagan test

data: anxiety1

BP = 1.1364, df = 2, p-value = 0.5666

```
#onset, offset, duration  
anxiety2 <- lm(  
  `Anxiety Rating` ~  
    `Sleep Duration` +  
    `Sleep Offset Time_Decimal Hour` +  
    `Sleep Onset Decimal Hour`,  
  data = step1  
)
```

```
summary(anxiety2)
```

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.1469	-0.7049	1.0838	1.3818	1.9541

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.0925675	1.0450930	7.743	1.66e-12 ***
`Sleep Duration`	0.0010575	0.0007641	1.384	0.169
`Sleep Offset Time_Decimal Hour`	0.0321888	0.1205281	0.267	0.790
`Sleep Onset Decimal Hour`	-0.0194257	0.0210236	-0.924	0.357

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.295 on 142 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.01977, Adjusted R-squared: -0.0009389

F-statistic: 0.9547 on 3 and 142 DF, p-value: 0.4161

```
bptest(anxiety2)
```

studentized Breusch-Pagan test

data: anxiety2

BP = 9.4659, df = 3, p-value = 0.0237

```
#onset, offset, duration, variability  
anxiety3 <- lm(  
  `Anxiety Rating` ~  
    `Sleep Duration` +  
    `Sleep Offset Time_Decimal Hour` +  
    `Sleep Onset Decimal Hour` +  
    onset_var +
```

```

    offset_var,
    data = step1
)

summary(anxiety3)

```

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.9946	-0.6792	1.0401	1.4167	2.1795

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.3598911	1.1990470	6.972	1.18e-10 ***
`Sleep Duration`	0.0011577	0.0007859	1.473	0.143
`Sleep Offset Time_Decimal Hour`	0.0354732	0.1231159	0.288	0.774
`Sleep Onset Decimal Hour`	-0.0248727	0.0223228	-1.114	0.267
onset_var	-0.0309295	0.0420798	-0.735	0.464
offset_var	-0.0512598	0.2551679	-0.201	0.841

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.318 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.02573, Adjusted R-squared: -0.009565

F-statistic: 0.729 on 5 and 138 DF, p-value: 0.6028

```
bptest(anxiety3)
```

studentized Breusch-Pagan test

data: anxiety3

BP = 10.383, df = 5, p-value = 0.06509

```
#Sleep Quality Analysis----
```

```
#onset & offset
```

```
sleep_quality1 <- lm(  
  `Sleep Quality` ~  
    `Sleep Offset Time_Decimal Hour` +  
    `Sleep Onset Decimal Hour`,  
  data = step1  
)
```

```
summary(sleep_quality1)
```

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.4794	-1.0730	0.3397	1.4783	3.0204

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.78792	0.92339	6.268	4.06e-09 ***
`Sleep Offset Time_Decimal Hour`	0.14209	0.10286	1.381	0.1693
`Sleep Onset Decimal Hour`	0.03639	0.01751	2.078	0.0395 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.03 on 143 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.03344, Adjusted R-squared: 0.01992

F-statistic: 2.473 on 2 and 143 DF, p-value: 0.0879

```
bptest(sleep_quality1)
```

studentized Breusch-Pagan test

data: sleep_quality1

BP = 1.2834, df = 2, p-value = 0.5264

```
#onset, offset, duration
sleep_quality2 <- lm(
  `Sleep Quality` ~
    `Sleep Duration` +
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`,
  data = step1
)

summary(sleep_quality2)
```

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour`, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.4114	-1.0942	0.2993	1.4779	3.0184

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.7779106	0.9269421	6.233	4.91e-09 ***
`Sleep Duration`	0.0002020	0.0006718	0.301	0.7641
`Sleep Offset Time_Decimal Hour`	0.1333790	0.1071760	1.244	0.2154
`Sleep Onset Decimal Hour`	0.0344898	0.0186702	1.847	0.0668 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.037 on 142 degrees of freedom

(26 observations deleted due to missingness)

Multiple R-squared: 0.03405, Adjusted R-squared: 0.01364

F-statistic: 1.669 on 3 and 142 DF, p-value: 0.1765

```
bptest(sleep_quality2)
```

studentized Breusch-Pagan test

data: sleep_quality2

BP = 4.0417, df = 3, p-value = 0.257

```
#onset, offset, duration, variability
sleep_quality3 <- lm(
  `Sleep Quality` ~
    `Sleep Duration` +
    `Sleep Offset Time_Decimal Hour` +
    `Sleep Onset Decimal Hour` +
    onset_var +
    offset_var,
  data = step1
)

summary(sleep_quality3)
```

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.4510	-1.0031	0.3062	1.4986	3.0138

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.8921655	1.0638612	5.538	1.49e-07 ***
`Sleep Duration`	0.0002554	0.0006951	0.367	0.714
`Sleep Offset Time_Decimal Hour`	0.1343439	0.1099508	1.222	0.224
`Sleep Onset Decimal Hour`	0.0320627	0.0199638	1.606	0.111
onset_var	-0.0102492	0.0376939	-0.272	0.786
offset_var	-0.0367834	0.2268967	-0.162	0.871

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.064 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.03469, Adjusted R-squared: -0.0002878

F-statistic: 0.9918 on 5 and 138 DF, p-value: 0.4252

```
bptest(sleep_quality3)
```

studentized Breusch-Pagan test

data: sleep_quality3

BP = 13.291, df = 5, p-value = 0.0208

```
#plots----
```

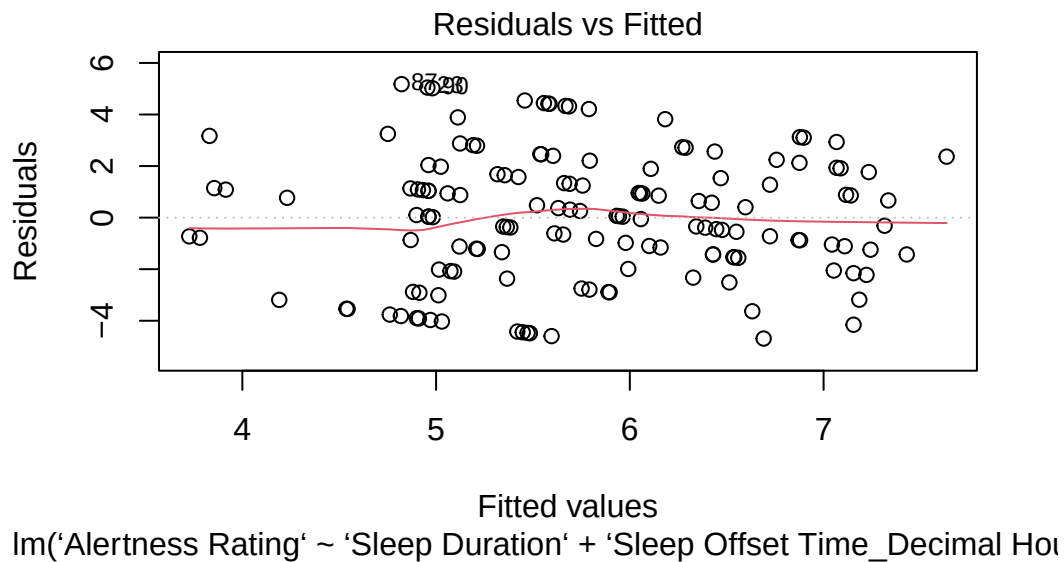
```
#CHECK MULTICOLLINEARITY
```

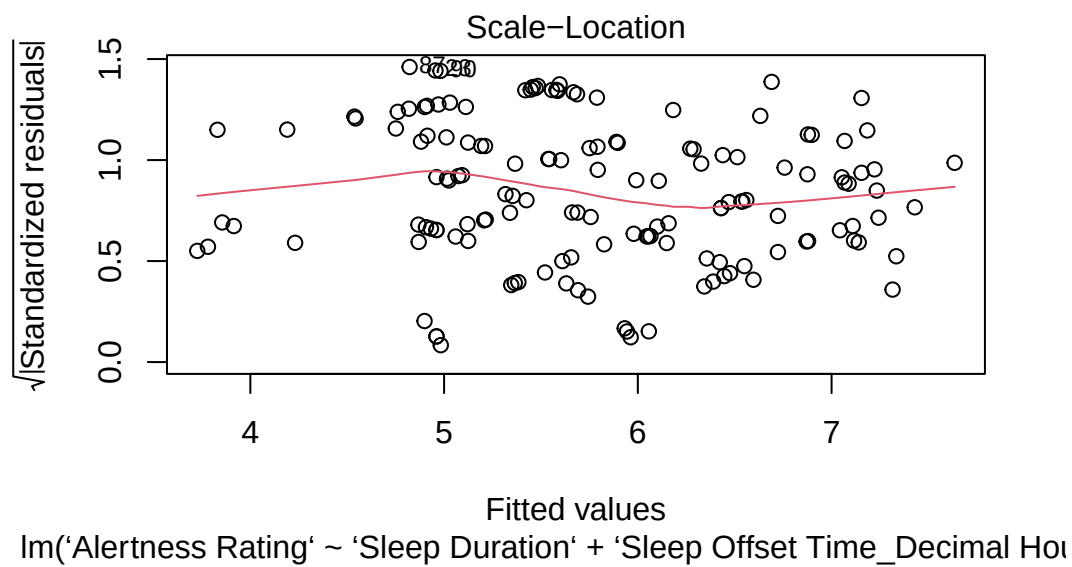
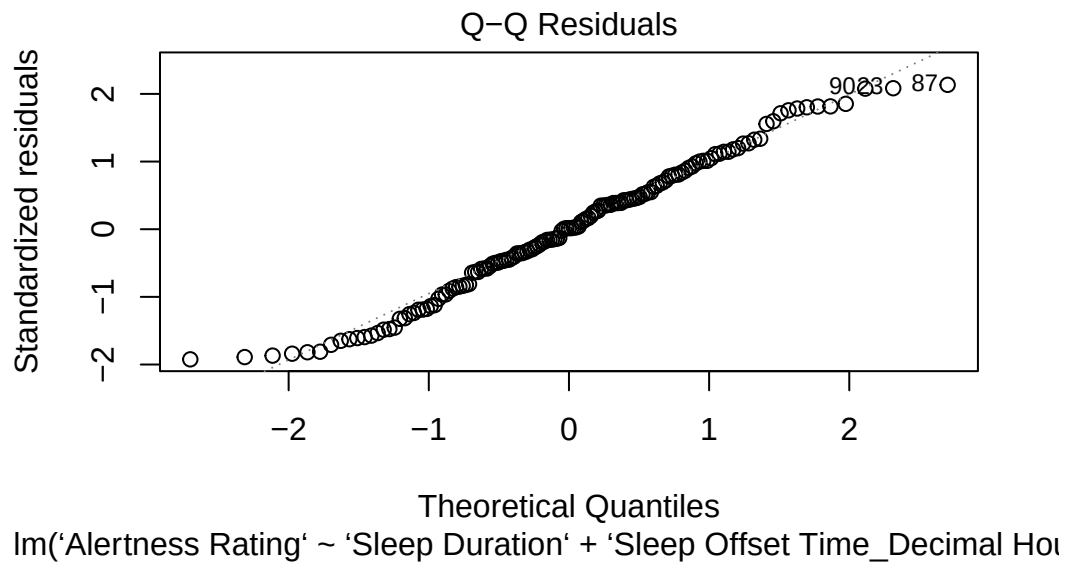
```
vif(alertness3)
```

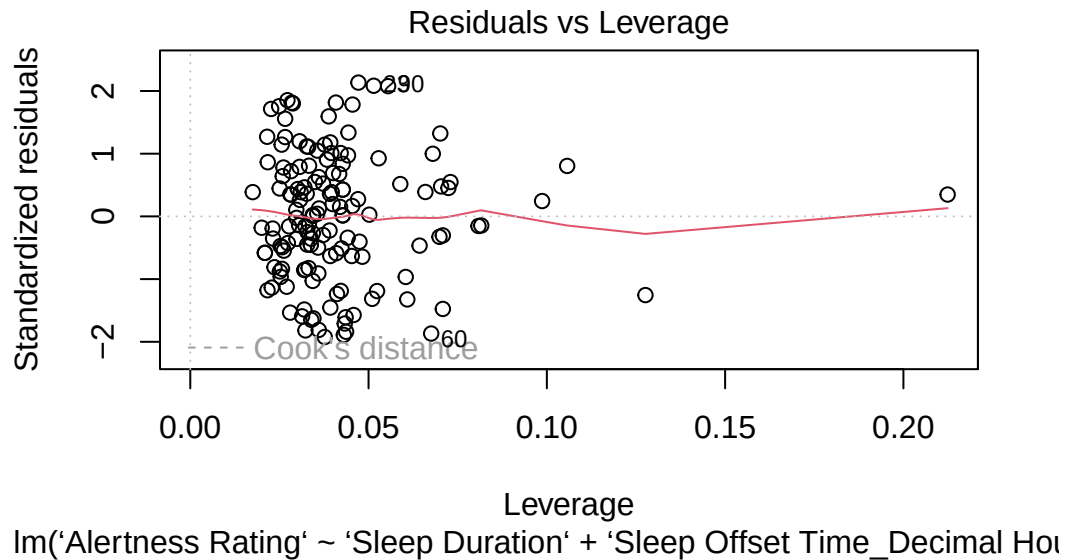
`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.204114	1.208512
`Sleep Onset Decimal Hour`	onset_var
1.376575	1.053563
offset_var	
1.078269	

```
#REGRESSION DIAGNOSTIC PLOTS
```

```
plot(alertness3)
```



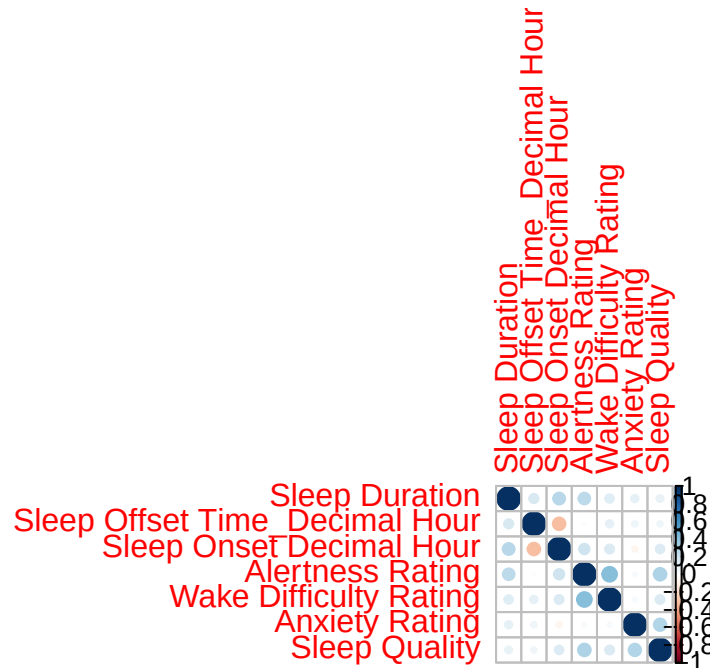




```
#CORRELATION MATRIX
```

```
vars <- step1 %>%
  select(
    `Sleep Duration`,
    `Sleep Offset Time_Decimal Hour`,
    `Sleep Onset Decimal Hour`,
    `Alertness Rating`,
    `Wake Difficulty Rating`,
    `Anxiety Rating`,
    `Sleep Quality`
  )

corrplot(cor(vars, use = "pairwise.complete.obs"),
  method = "circle")
```



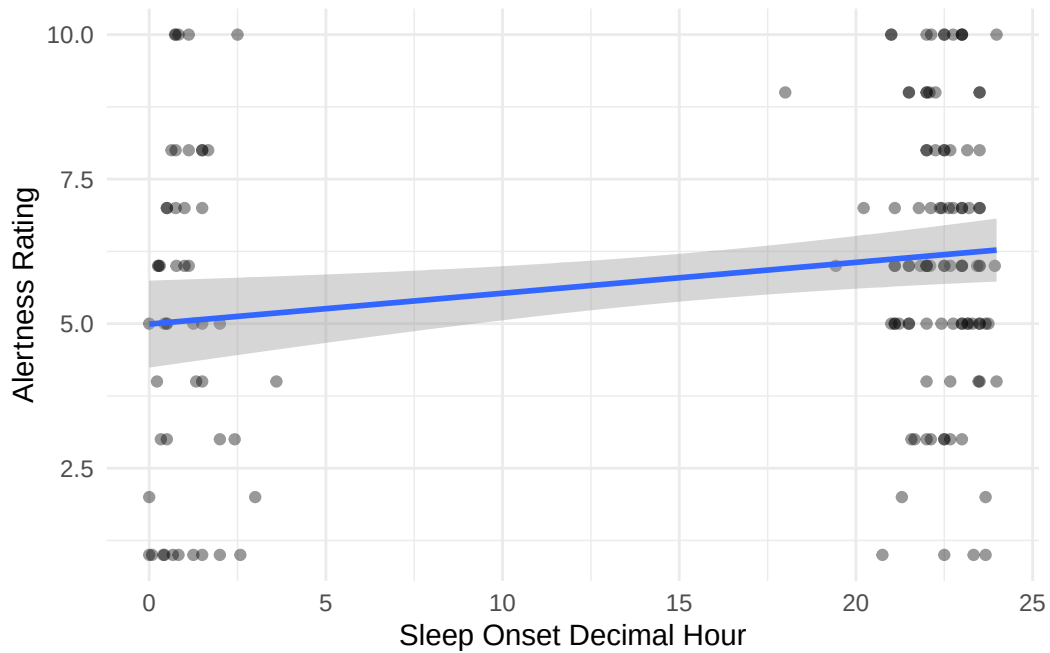
```
#SCATTERPLOTS WITH REGRESSION LINES
```

```
ggplot(step1,
  aes(x = `Sleep Onset Decimal Hour`,
    y = `Alertness Rating`)) +
  geom_point(alpha = .4) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 24 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 24 rows containing missing values or values outside the scale range
(`geom_point()`).
```

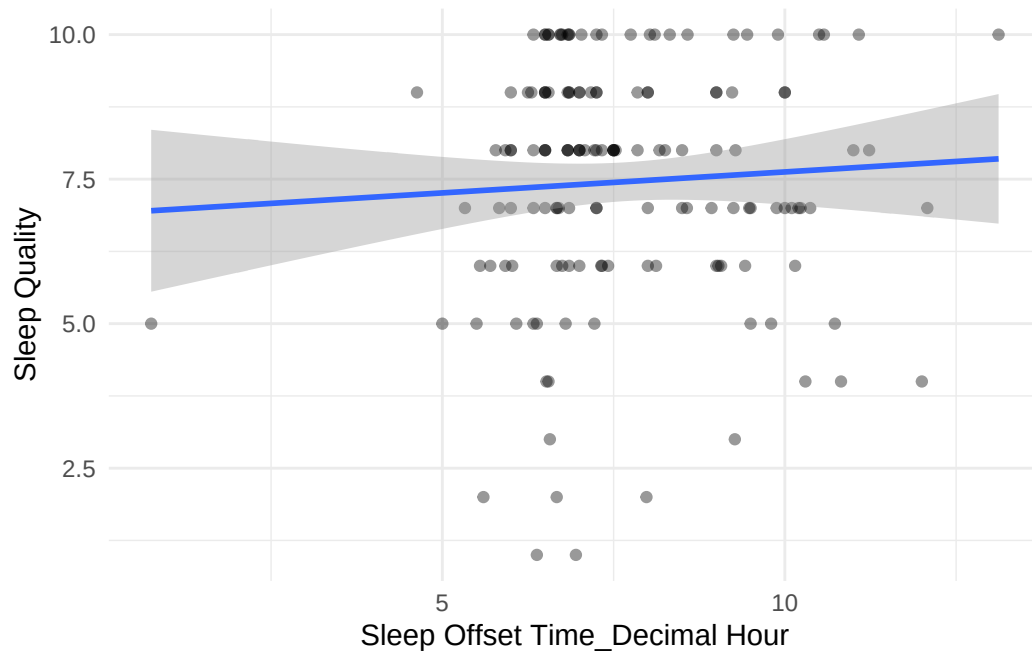


```
ggplot(step1,
  aes(x = `Sleep Offset Time_Decimal Hour`,
    y = `Sleep Quality`)) +
  geom_point(alpha = .4) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 25 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 25 rows containing missing values or values outside the scale range
(`geom\_point()`).



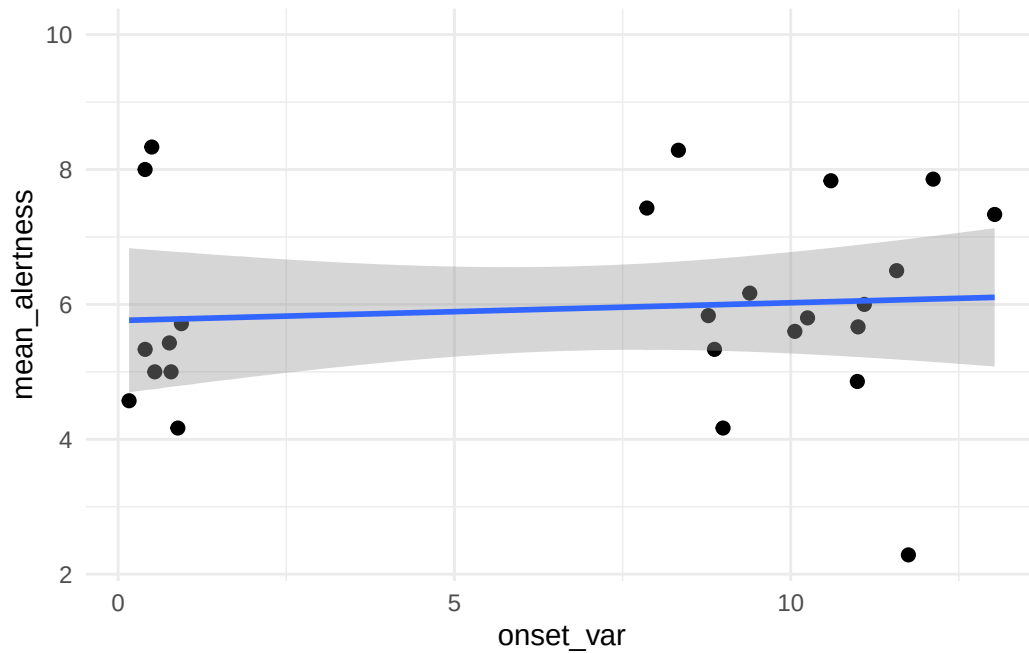
```
#VARIABILITY EFFECTS VISUALIZATION
```

```
ggplot(sleep_var,
       aes(x = onset_var,
           y = mean_alertness)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

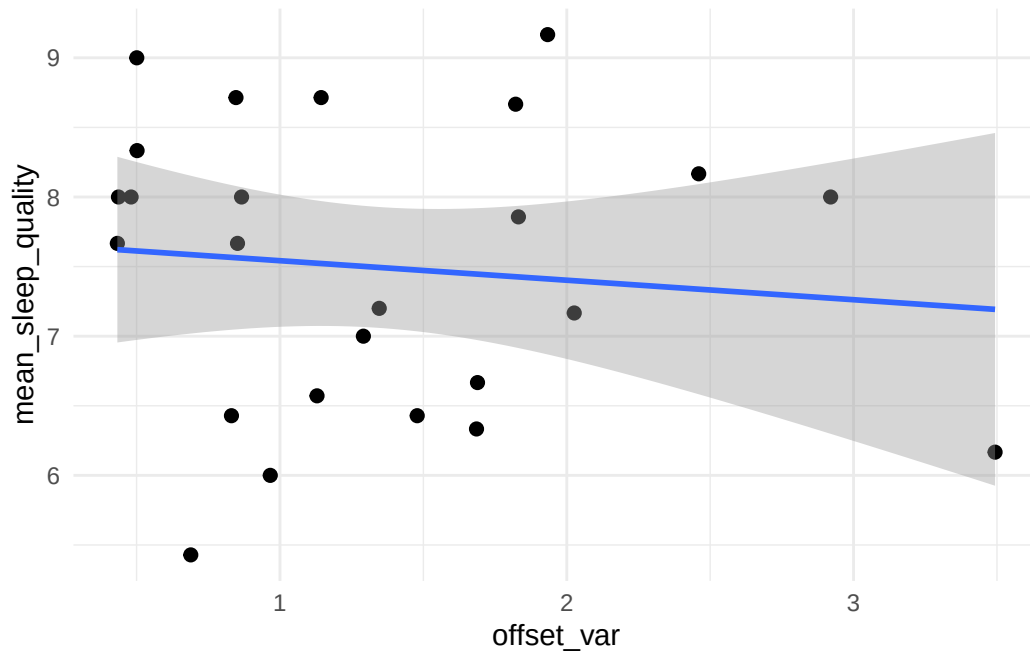
```
Warning: Removed 3 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```
ggplot(sleep_var,
       aes(x = offset_var,
           y = mean_sleep_quality)) +
  geom_point(size = 2) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range (`stat\_smooth()`).
 Removed 3 rows containing missing values or values outside the scale range
 (`geom\_point()`).



```
#Pearson Correlation on DV-----
#install.packages("Hmisc")
#install.packages("corrplot")
#install.packages("GGally")
library(Hmisc)
library(corrplot)
library(GGally)

#dependent variables
dep_vars <- step1[, c(
  "Alertness Rating",
  "Wake Difficulty Rating",
  "Anxiety Rating",
  "Sleep Quality"
)]

#Pearson correlation matrix
cor_matrix <- cor(dep_vars,
  use = "pairwise.complete.obs",
  method = "pearson")

print(cor_matrix)
```

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	1.00000000	0.42018413	0.03308203
Wake Difficulty Rating	0.42018413	1.00000000	0.03607769
Anxiety Rating	0.03308203	0.03607769	1.00000000
Sleep Quality	0.30883513	0.15515198	0.29848043

	Sleep Quality
Alertness Rating	0.3088351
Wake Difficulty Rating	0.1551520
Anxiety Rating	0.2984804
Sleep Quality	1.0000000

```
#Correlations WITH p-values
cor_test_results <- rcorr(as.matrix(dep_vars),
                           type = "pearson")

print(cor_test_results)
```

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	1.00	0.42	0.03
Wake Difficulty Rating	0.42	1.00	0.04
Anxiety Rating	0.03	0.04	1.00
Sleep Quality	0.31	0.16	0.30

	Sleep Quality
Alertness Rating	0.31
Wake Difficulty Rating	0.16
Anxiety Rating	0.30
Sleep Quality	1.00

n

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
Alertness Rating	149	146	148
Wake Difficulty Rating	146	146	145
Anxiety Rating	148	145	148
Sleep Quality	148	145	147

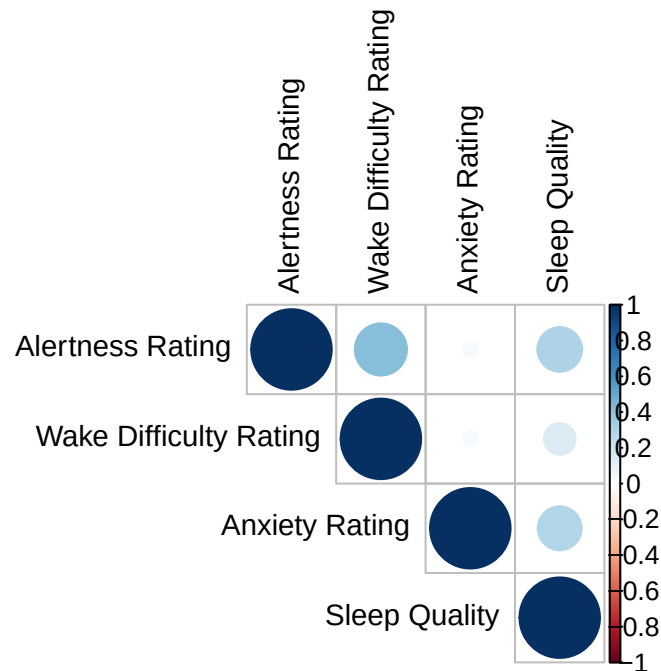
	Sleep Quality
Alertness Rating	148
Wake Difficulty Rating	145
Anxiety Rating	147
Sleep Quality	148

P

	Alertness Rating	Wake Difficulty Rating	Anxiety Rating
--	------------------	------------------------	----------------

Alertness Rating		0.0000	0.6898
Wake Difficulty Rating	0.0000		0.6666
Anxiety Rating	0.6898	0.6666	
Sleep Quality	0.0001	0.0624	0.0002
	Sleep Quality		
Alertness Rating	0.0001		
Wake Difficulty Rating	0.0624		
Anxiety Rating	0.0002		
Sleep Quality			

```
#Correlation plot
corrplot(cor_matrix,
  method = "circle",
  type = "upper",
  tl.col = "black",
  tl.cex = 0.9)
```



```
#Scatterplot matrix with regression lines
ggpairs(dep_vars)
```

Warning: Removed 23 rows containing non-finite outside the scale range (`stat\_density()`).

Warning in ggally_statistic(data = data, mapping = mapping, na.rm = na.rm, :
Removed 26 rows containing missing values

Warning in ggally_statistic(data = data, mapping = mapping, na.rm = na.rm, :
Removed 24 rows containing missing values

Warning in ggally_statistic(data = data, mapping = mapping, na.rm = na.rm, :
Removed 24 rows containing missing values

Warning: Removed 26 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 26 rows containing non-finite outside the scale range
(`stat\_density()`).

Warning in ggally_statistic(data = data, mapping = mapping, na.rm = na.rm, :
Removed 27 rows containing missing values

Warning in ggally_statistic(data = data, mapping = mapping, na.rm = na.rm, :
Removed 27 rows containing missing values

Warning: Removed 24 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 27 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 24 rows containing non-finite outside the scale range
(`stat\_density()`).

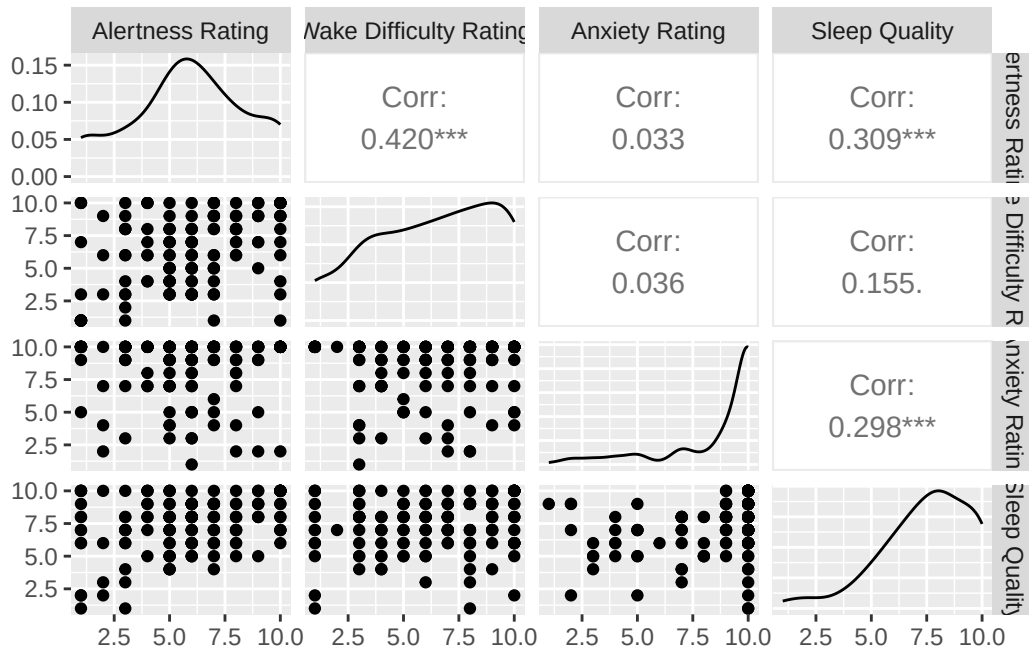
Warning in ggally_statistic(data = data, mapping = mapping, na.rm = na.rm, :
Removed 25 rows containing missing values

Warning: Removed 24 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 27 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 25 rows containing missing values or values outside the scale range
(`geom\_point()`).

Warning: Removed 24 rows containing non-finite outside the scale range
(`stat\_density()`).



```
#running robust standard errors----
library(sandwich)
library(lmtest)

coeftest(alertness3, vcov = vcovHC(alertness3, type = "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.71750126	1.20380995	3.0881	0.002433	**
`Sleep Duration`	0.00227718	0.00087143	2.6131	0.009959	**
`Sleep Offset Time_Decimal Hour`	0.03539554	0.13325336	0.2656	0.790921	
`Sleep Onset Decimal Hour`	0.03825066	0.02602504	1.4698	0.143886	
onset_var	0.05858558	0.05074411	1.1545	0.250264	
offset_var	-0.28210566	0.22575268	-1.2496	0.213537	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
coeftest(anxiety3, vcov = vcovHC(anxiety3, type = "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.35989111	1.67780188	4.9826	1.847e-06 ***
`Sleep Duration`	0.00115773	0.00086787	1.3340	0.1844
`Sleep Offset Time_Decimal Hour`	0.03547316	0.17240382	0.2058	0.8373
`Sleep Onset Decimal Hour`	-0.02487272	0.02642234	-0.9414	0.3482
onset_var	-0.03092954	0.04037443	-0.7661	0.4449
offset_var	-0.05125984	0.32376833	-0.1583	0.8744

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
coeftest(wake_diff3, vcov = vcovHC(wake_diff3, type = "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.99234644	1.32378744	3.7713	0.0002414 ***
`Sleep Duration`	0.00112287	0.00087052	1.2899	0.1992770
`Sleep Offset Time_Decimal Hour`	0.21709630	0.14366658	1.5111	0.1330800
`Sleep Onset Decimal Hour`	0.03100672	0.02764274	1.1217	0.2639687
onset_var	-0.06414519	0.05279502	-1.2150	0.2264768
offset_var	-0.71964942	0.25775204	-2.7920	0.0059928 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
coeftest(sleep_quality3, vcov = vcovHC(sleep_quality3, type = "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.89216552	1.11874964	5.2667	5.206e-07 ***
`Sleep Duration`	0.00025542	0.00070248	0.3636	0.7167
`Sleep Offset Time_Decimal Hour`	0.13434388	0.11047080	1.2161	0.2260
`Sleep Onset Decimal Hour`	0.03206272	0.02098630	1.5278	0.1289

onset_var	-0.01024923	0.04173218	-0.2456	0.8064
offset_var	-0.03678338	0.18998515	-0.1936	0.8468

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# graphs for the final presentation----

#install.packages("ggplot2")
#install.packages("corrplot")
#install.packages("broom")
#install.packages("dplyr")

library(ggplot2)
library(corrplot)
library(broom)
library(dplyr)

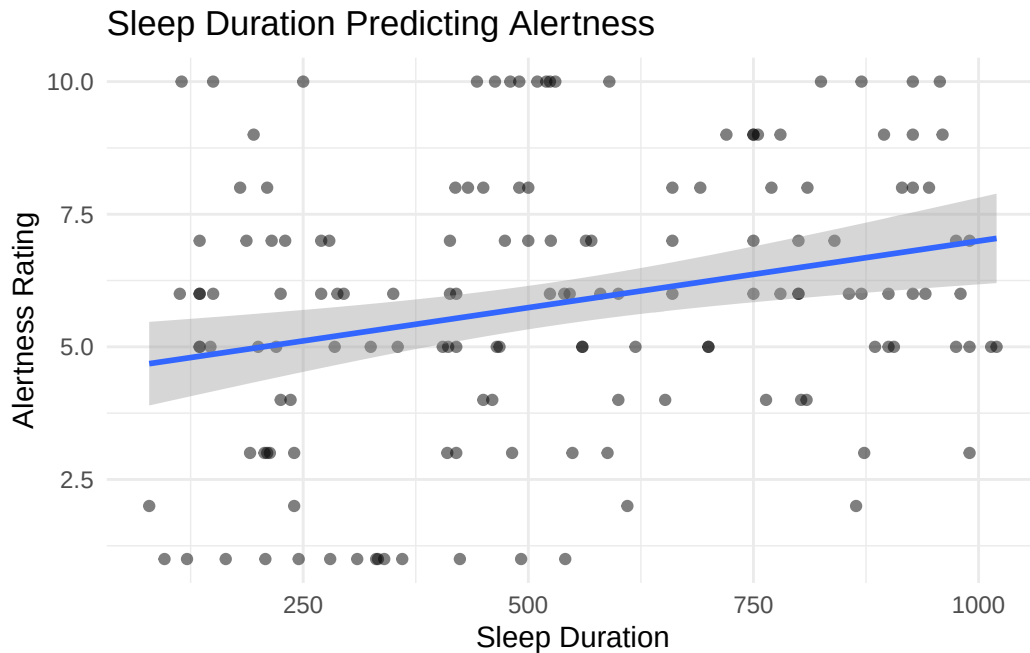
# GRAPH 1: Sleep Duration → Alertness

ggplot(step1,
       aes(x = `Sleep Duration`,
           y = `Alertness Rating`)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() +
  labs(title = "Sleep Duration Predicting Alertness",
       x = "Sleep Duration",
       y = "Alertness Rating")
```

`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 24 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 24 rows containing missing values or values outside the scale range
(`geom\_point()`).



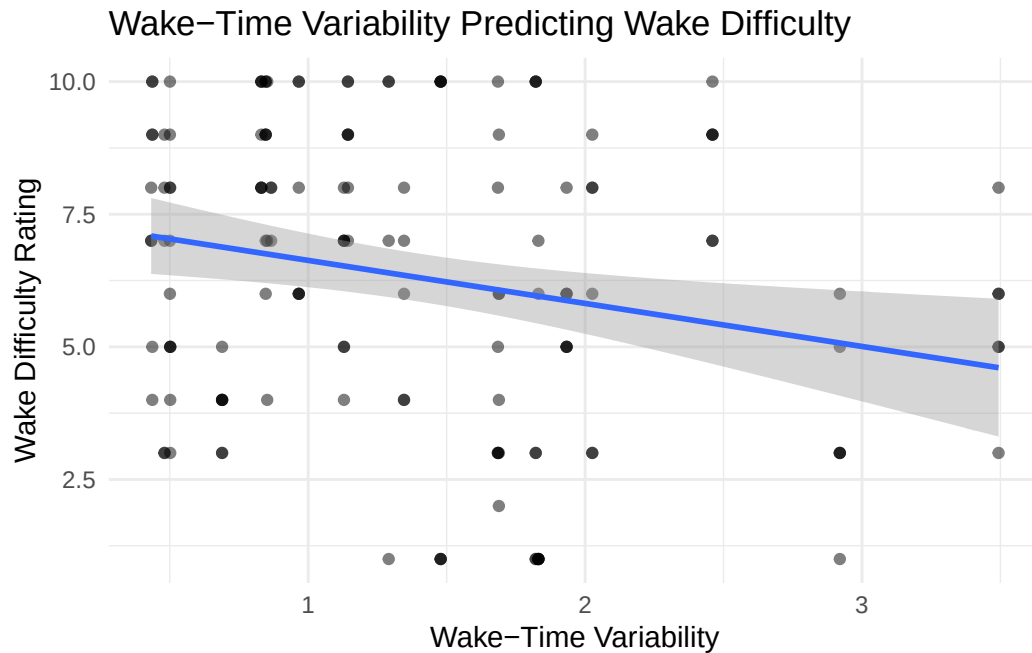
```
# GRAPH 2: Offset Variability → Wake Difficulty
```

```
ggplot(step1,
  aes(x = offset_var,
      y = `Wake Difficulty Rating`)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() +
  labs(title = "Wake-Time Variability Predicting Wake Difficulty",
       x = "Wake-Time Variability",
       y = "Wake Difficulty Rating")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 28 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 28 rows containing missing values or values outside the scale range
(`geom_point()`).
```



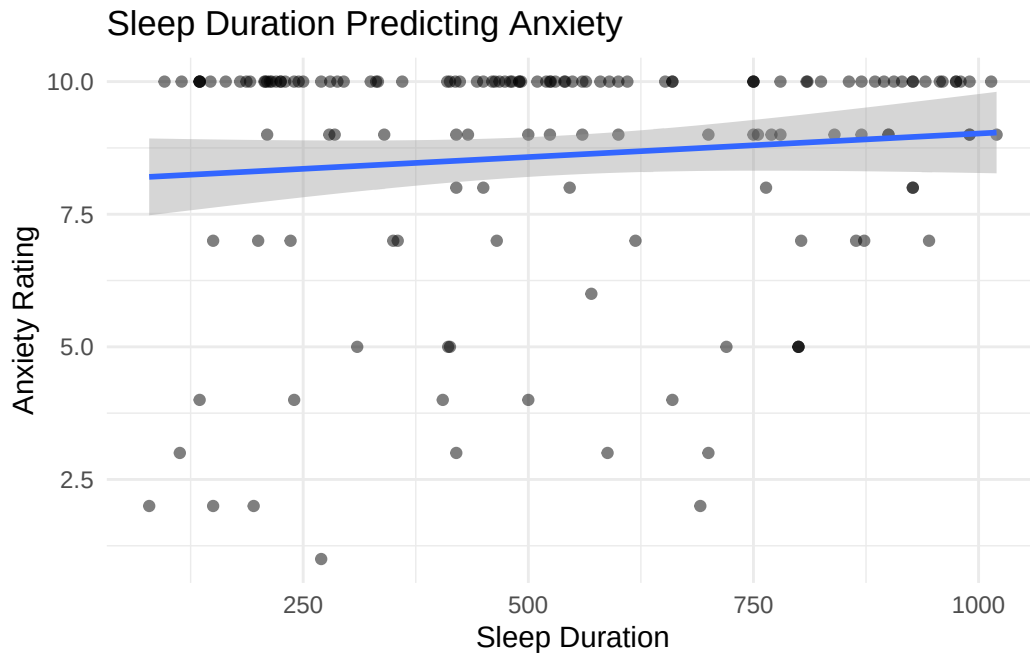
GRAPH 3: Duration → Anxiety

```
ggplot(step1,
  aes(x = `Sleep Duration`,
    y = `Anxiety Rating`)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() +
  labs(title = "Sleep Duration Predicting Anxiety",
    x = "Sleep Duration",
    y = "Anxiety Rating")
```

``geom_smooth()`` using formula = `'y ~ x'`

Warning: Removed 25 rows containing non-finite outside the scale range (``stat_smooth()``).

Warning: Removed 25 rows containing missing values or values outside the scale range (``geom_point()``).



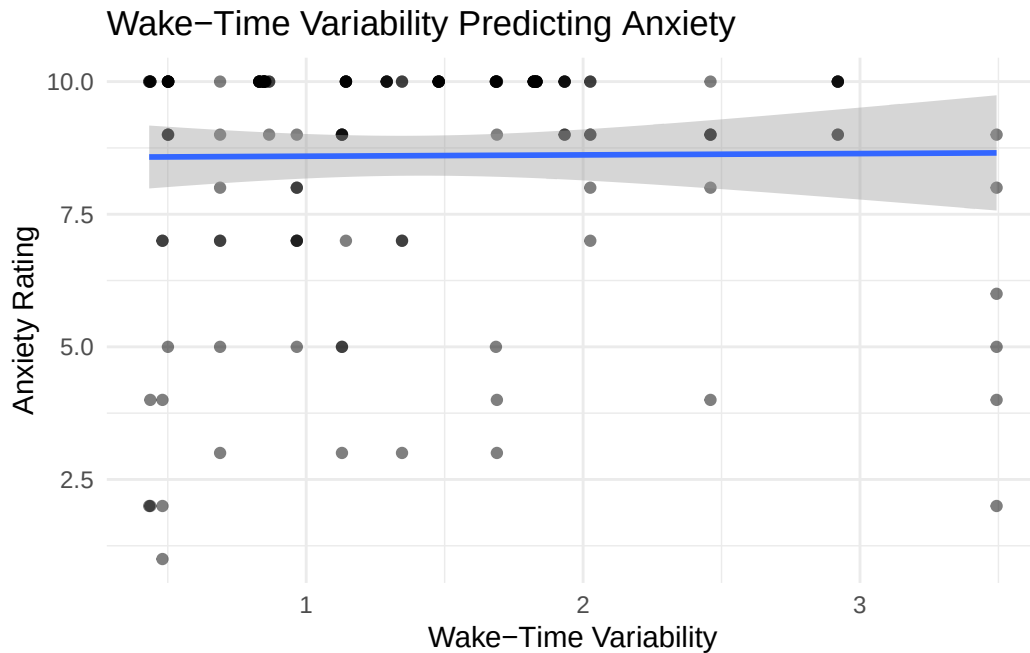
GRAPH 4: Offset Variability → Anxiety

```
ggplot(step1,
  aes(x = offset_var,
      y = `Anxiety Rating`)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() +
  labs(title = "Wake-Time Variability Predicting Anxiety",
       x = "Wake-Time Variability",
       y = "Anxiety Rating")
```

``geom_smooth()`` using formula = 'y ~ x'

Warning: Removed 26 rows containing non-finite outside the scale range (``stat_smooth()``).

Warning: Removed 26 rows containing missing values or values outside the scale range (``geom_point()``).



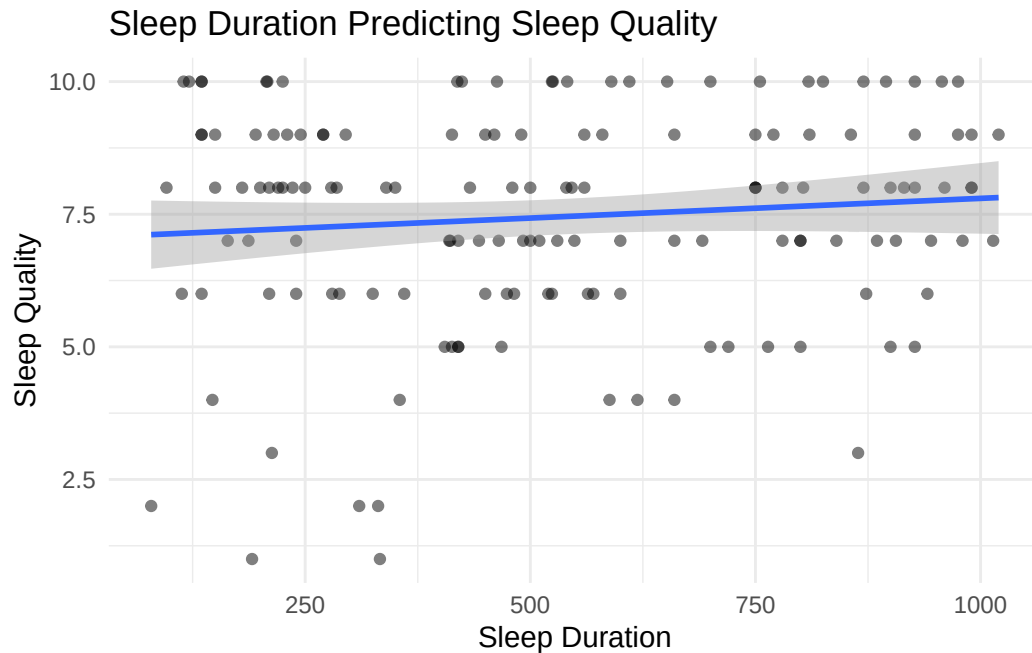
```
# GRAPH 5: Duration → Sleep Quality
```

```
ggplot(step1,
  aes(x = `Sleep Duration`,
    y = `Sleep Quality`)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() +
  labs(title = "Sleep Duration Predicting Sleep Quality",
    x = "Sleep Duration",
    y = "Sleep Quality")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 25 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 25 rows containing missing values or values outside the scale range
(`geom_point()`).
```

GRAPH 6: Offset Time → Sleep Quality

```
ggplot(step1,
  aes(x = `Sleep Offset Time_Decimal Hour`,
    y = `Sleep Quality`)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm", se = TRUE) +
  theme_minimal() +
  labs(title = "Wake Time Predicting Sleep Quality",
    x = "Sleep Offset (Decimal Hour)",
    y = "Sleep Quality")
```

``geom_smooth()`` using formula = `'y ~ x'`

Warning: Removed 25 rows containing non-finite outside the scale range (``stat_smooth()``).

Removed 25 rows containing missing values or values outside the scale range (``geom_point()``).

